

Ê PHASOR TEMPLATES (Air: ε_r=μ_r=1)

Wavenumber per medium <i>nonmag.</i>			
$k_i = \frac{\omega\sqrt{\epsilon_{r,i}}}{c}$ (rad/m) $\Rightarrow k_2 = k_1\sqrt{\epsilon_{r,2}/\epsilon_{r,1}}$			
Direction table $u \in \{x, y, z\}$ = axis perp. to boundary			
Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

+û direction ⇔ Med 2 at u ≥ 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. ê, amplitude a ₀):		
$\hat{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u}$ (V/m)	(incident wave, Med 1)	
$\hat{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u}$ (V/m)	(reflected wave, Med 1)	
$\hat{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u}$ (V/m)	(transmitted wave, Med 2)	
(lossy med 2: replace $e^{\mp j k_2 u}$ with $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$)		
Total in medium 1: $\hat{\mathbf{E}}_1 = \hat{\mathbf{E}}^i + \hat{\mathbf{E}}^r$.		
ê by polarization plug into templates above		
Pol.	ê	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on ê unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

<i>Med. 1 (z < 0) incident side; Med. 2 (z ≥ 0) transmitted.</i>			
	⊥ pol (θ _i = 0)	pol	
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{ } = (1 + \Gamma_{ }) \frac{\cos \theta_i}{\cos \theta_t}$
R refl. pwr	$ \Gamma ^2$	$ \Gamma_{\perp} ^2$	$ \Gamma_{ } ^2$
T trans. pwr	$ \tau ^2 \frac{\eta_2}{\eta_1}$	$ \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$	$ \tau_{ } ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
R+T total	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{ } = 1 - R_{ }$
Notes: $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; <i>nonmag.</i> $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$.			
Intrinsic impedance η_i plug into table above			
$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$			
Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says “ferromagnetic”/iron/ferrite/etc.			
Low-loss correction $\eta_c \propto \sigma / (\omega \epsilon) \ll 1$			
$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega \epsilon} \right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$			
For Γ/τ just use $\eta_0 / \sqrt{\epsilon_r}$. The j term is for $ \eta_c , \theta_{\eta}$. See LOSSY MEDIA — 5 CASES.			

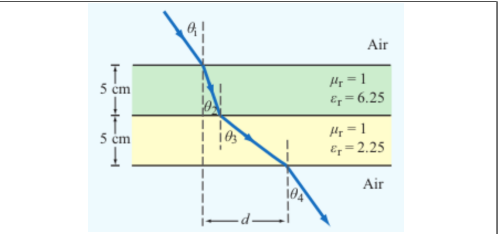
POWER REFLECTION & TRANSMISSION

% reflected <i>always</i> %P _r = 100 Γ ²
% transmitted <i>η-ratio matters!</i> %P _t = 100 τ ² η ₁ /η ₂
Check: %P _r + %P _t = 100.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary <i>angles from normal</i>	$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$
Multilayer chain <i>stack of slabs</i> 1 → 2 → 3 → 4	$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$
Air-slabs-air shortcut <i>outer media identical</i>	$\theta_{\text{exit}} = \theta_{\text{incident}}$
<i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i>	

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i Slab-indexed θ_i = angle inside slab i; t_i = slab thickness (m)

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent $\epsilon' = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.854 \times 10^{-12}$ F/m		
$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$		
Type	$\sigma / \omega \epsilon$	η_c, α, β
Lossless (σ=0)	0	$\eta = \eta_0 / \sqrt{\epsilon_r}$ exact $\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left(1 + j \frac{\sigma \epsilon_r}{2\omega \epsilon} \right)$ $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$
Quasi-cond.	$10^{-2} - 10^2$	exact: $(\eta / \sqrt{X}) e^{j\theta_{\eta}}$ see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j) \sqrt{\pi f \mu / \sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
Magnetic	any	keep full $\sqrt{\mu_r / \epsilon_r} \eta_0$

For Γ/τ: drop j correction; use $\eta_0 / \sqrt{\epsilon_r}$. Magnetic: keep μ_r.

Low-loss quick params $\sigma / (\omega \epsilon) \ll 1, \mu_r = 1$
 $\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}$ (Np/m), $\beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$ (rad/m), $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}}$ (Ω)
Exact (quasi-cond) $X = \sqrt{1 + (\epsilon'' / \epsilon')^2}$
 $\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X - 1}, \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X + 1}$
 $\theta_{\eta} = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$ TM mode $H_z = 0, E_z \neq 0$
Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.
E_x form (TE) read m, n from arguments
 $E_x \propto \cos \left(\frac{m\pi x}{a} \right) \sin \left(\frac{n\pi y}{b} \right) \sin(\omega t - \beta z)$
coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If m=0 or n=0 in the mode label ⇒ must be TE (TM forbids 0).
3. Source field H_z(x, y) given ⇒ TE. Source field E_z(x, y) given ⇒ TM.

TE allows at least one of m, n ≥ 1 (other can be 0). TM needs both m, n ≥ 1. So TE₁₀ exists but TM₁₀ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m, $\tilde{H}$ in A/m, Z in Ω.

TE mode <i>generator: $\tilde{H}_z$</i>	
$\tilde{H}_z = H_0 \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b} \right) H_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a} \right) H_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{E}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{TE}, \tilde{H}_y = \tilde{E}_x / Z_{TE}$	
$Z_{TE} = \eta / \sqrt{1 - (f_c / f)^2}$	
TM mode <i>generator: $\tilde{E}_z$</i>	
$\tilde{E}_z = E_0 \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a} \right) E_0 \cos \frac{m\pi x}{a} \sin \frac{n\pi y}{b}$	
$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b} \right) E_0 \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$	
$\tilde{H}_z = 0, \tilde{H}_x = -\tilde{E}_y / Z_{TM}, \tilde{H}_y = \tilde{E}_x / Z_{TM}$	
$Z_{TM} = \eta \sqrt{1 - (f_c / f)^2}$	
TEM plane wave <i>for reference</i>	
$\tilde{E}_x = E_0 e^{-j\beta z}, \tilde{H}_y = \tilde{E}_x / \eta$	
$\eta = \sqrt{\mu / \epsilon} \text{ (}\Omega\text{)}, f_c \text{ N/A}, k = \omega \sqrt{\mu \epsilon}$	
Properties common to TE/TM	
$f_c = \frac{u_{p0}}{2} \sqrt{(m/a)^2 + (n/b)^2} \text{ (Hz)}$	
$\beta = k \sqrt{1 - (f_c / f)^2} \text{ (rad/m)}$	
$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1 - (f_c / f)^2}} \text{ (m/s)}$	

CUTOFF FREQUENCY

Common cutoffs (a > b) f _c formula in Table 8-3	
Mode	f _{mn}
u _{p0}	c / √ε _r (hollow: u _{p0} = c)
TE ₁₀	f ₁₀ = u _{p0} / (2a) (dominant)
TE ₀₁	f ₀₁ = u _{p0} / (2b)
TE ₂₀	f ₂₀ = u _{p0} / a = 2f ₁₀
TE ₁₁ , TM ₁₁	f ₁₁ = $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires f > f_{mn}. In Z_{TE}/Z_{TM} and u_g, f_c = f_{mn} of the excited mode.

ε_r FROM DISPERSION

Given ω, β, mode (m, n) result is unitless	
$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 \right]$	(unitless)
Inputs: ω (rad/s), β (rad/m), a, b (m), c = 3 × 10 ⁸ (m/s), m, n (unitless).	

GROUP VELOCITY & TRAVEL TIME

$u_g = u_{p0} \sqrt{1 - (f_{mn} / f)^2}$ (m/s), $u_p u_g = u_{p0}^2$
 $t = L / u_g$ (s) travel time through guide of length L
Higher modes have lower u_g ⇒ arrive later.

BOUNDARY

η_i, θ_t — imp. (Ω)
Γ, τ, R, T — coeffs (unitless)
τ = 1 + Γ; R = |Γ|²
k_i = ω√ε_{r,i}/c (rad/m)
α₂ (Np/m), β₂ (rad/m): lossy
z = 0 — boundary plane (m)
η₀ = 120π ≈ 377 Ω

SNELL / OBLIQUE

θ_i, θ_t — angles from normal
n_i = √ε_{r,i} — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
⊥: **E** ⊥ this plane
||: **E** || this plane

WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
Z_{TE}/TM — wave imp. (Ω)
E_x — E-field comp. (V/m)
H_y — H-field comp. (A/m)
η = η₀ / √ε_r

VELOCITIES

u_{p0} = c / √ε_r — bulk phase vel.
u_p = u_{p0} / √(1 - (f_c/f)²)
u_g = u_{p0} √(1 - (f_c/f)²)
u_p u_g = u_{p0}²
t = L / u_g — travel time
c = 3 × 10⁸ m/s

POWER & UNITS

%P_r, %P_t — pwr refl./trans. (unitless)
%P_r = 100|Γ|²
%P_t = 100|τ|² η₁/η₂
%P_r + %P_t = 100
σ (S/m), ε (F/m), μ (H/m)
ε₀ = 8.85 × 10⁻¹² F/m
μ₀ = 4π × 10⁻⁷ H/m

DIPOLE TYPE BY LENGTH

STEP 0 — always start here $\lambda = \text{wavelength (m)}, l = \text{antenna rod length (m)}, c = 3 \times 10^8 \text{ m/s}$

$\lambda = \frac{c}{f} \text{ (m)}$	ratio $\frac{l}{\lambda} \rightarrow$	pick row below
l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta), D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D = 1.64, R_{\text{rad}} = 73 \Omega$
$= 1$	Full-wave	ARB, $D = 2.41, R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta, \quad \vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$
l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k = 2\pi/\lambda$ (rad/m); $\eta_0 = 120\pi \approx 377 \Omega$.
Max @ broadside $\theta_{\text{max}} = \pi/2$; D exact $D = 1.5$ (1.76 dB)
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2 \text{ (W)}$

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l
$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.
Normalized pattern <i>dimensionless</i> $F(\theta) = S(\theta)/S_{\text{max}}$
Quarter-wave $l = \lambda/4 \quad \pi l/\lambda = \pi/4, \cos(\pi/4) = \sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$
$D = 1.64, R_{\text{rad}} = 36.5 \Omega, \theta_{\text{max}} = \pi/2$.
Half-wave $l = \lambda/2 \quad \pi l/\lambda = \pi/2$, <i>shortcut form</i>
$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$
$D = 1.64, R_{\text{rad}} = 73 \Omega, \theta_{\text{max}} = \pi/2$.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion <i>multiply by $\pi/180$ or $180/\pi$</i>
$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180} \quad \theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$
$^\circ \quad \text{rad} \quad \sin \theta \quad \cos \theta \quad \sin^2 \theta \quad \cos^2 \theta$
0 0 0 1 0 1
30 $\pi/6$ 1/2 $\sqrt{3}/2$ 1/4 3/4
45 $\pi/4$ $\sqrt{2}/2$ $\sqrt{2}/2$ 1/2 1/2
60 $\pi/3$ $\sqrt{3}/2$ 1/2 3/4 1/4
90 $\pi/2$ 1 0 1 0
180 π 0 -1 0 1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Setup N identical elements along axis, spacing d
elem pos $z_i = id, i = 0, \dots, N-1; A_i = a_i e^{j\psi_i}$
Equal phase ($\psi_i = 0$) $\Rightarrow$ broadside beam ($\theta_{\text{max}} = \pi/2$).
Progressive phase $\psi_i = i\alpha \Rightarrow$ steered beam at $\cos \theta_{\text{max}} = -\alpha/(kd)$.
Pattern multiplication $S_e = \text{single element pattern}$
$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$
Array factor (general) $k = 2\pi/\lambda$
$F_a(\theta) = \left \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right ^2$
Uniform $A_i = 1$, equal phase <i>closed form</i>
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \quad \gamma = kd \cos \theta$
Peak: $F_a^{\text{max}} = N^2$ at $\gamma = 0$ (broadside, $\theta = \pi/2$). Normal-ize: $F_a^{\text{norm}} = F_a/N^2$.
HPBW (broadside, uniform) <i>use Nd, not $(N-1)d$</i>
$\beta \approx 0.88 \frac{\lambda}{Nd} \text{ (rad)} = 50.8^\circ \cdot \frac{\lambda}{Nd}$
Grating lobes appear if $d > \lambda$ (broadside). Avoid.

DIPOLE

l — antenna length (m)
$\lambda = c/f$ (m); $c = 3 \times 10^8 \text{ m/s}$
I_0 — peak current (A)
$k = 2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0 = 120\pi \approx 377 \Omega$
$S(R, \theta)$ — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
a — coef. on θ^2 in Gaussian
I_0 — HPBW (rad or $^\circ$)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity
ξ — rad. efficiency; $G = \xi D$
$D_{\text{dB}} = 10 \log_{10} D; D = 10^{D_{\text{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
P_r — power received (W)
G_t — transmit gain (linear)
G_r — receive gain (linear)
$S_r = G_t P_t / (4\pi R^2)$ (W/m ²)
$P_r = G_t G_r (\lambda / 4\pi R)^2 P_t$
$G = 10^{G_{\text{dB}}/10}$
3 dB = $\times 2, 10 \text{ dB} = \times 10$
$P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m ²)
$A_e = \lambda^2 D / (4\pi)$ (m ²)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p / \lambda^2$ (unitless)
$\beta \approx 0.88 \lambda / \ell$ (rect/sq.)
$\beta \approx 1.02 \lambda / d$ (circular)
$2A_p \Rightarrow 2D, \beta/\sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma = kd \cos \theta$
$F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
$F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
$F_a^{\text{norm}} = F_a/N^2$
$\beta \approx 0.88 \lambda / (Nd)$ (broadside)

BEAM PARAMETERS — β, Ω_p, D

Normalize <i>always start here</i>
$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)
Beamwidth β at threshold X
Set $F(\theta_X) = X$, solve for $\theta_X; \beta = 2\theta_X$ (symmetric)
Threshold $X = 10^{\text{dB}/10} \quad \theta_X$ (Gauss. short-cut)
HPBW (−3 dB) 0.5 $\sqrt{\frac{\ln 2}{a}}$
−6 dB 0.25 $\sqrt{\frac{\ln 4}{a}}$
−10 dB 0.1 $\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$ any $X \quad F(\theta_X) = X$
Pattern solid angle Ω_p <i>recognize pattern, look up</i>
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi$ (sr)
No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta d\theta$.
θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.
Pattern $F(\theta)$ Ω_p (sr)
Isotropic ($F = 1$) $4\pi \approx 12.57$
Hertzian $\sin^2 \theta$ $8\pi/3 \approx 8.38$
Half-wave dipole ≈ 7.66
Narrow Gauss $e^{-a\theta^2}$ $\pi/a = \pi\theta_{1/2}^2 / \ln 2$
2-axis HPBW (aperture) $\beta_{xz} \cdot \beta_{yz}$
Other (use integral) numerical

TI-36X Pro: [2nd] [d/dx] for $\int_{\square}^{\square}, \text{RAD mode, multiply by } 2\pi.$

Directivity D <i>always $D = 4\pi/\Omega_p$; common shortcuts below</i>
$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$
WORD TRAP: “direction of max radiation” = angle θ_{max} (rad/deg). “directivity” = unitless number D . Different things!
Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p / \lambda^2$. 2-axis HPBW: $D \approx 4\pi / (\beta_{xz} \beta_{yz})$. Circular: $D \approx 4\pi / \beta^2$.
Gain (if asked) $\xi = \text{rad. eff.}, 0 < \xi \leq 1$
$G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G = D$.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2 \text{ W}$. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.
Lossless antenna ($\xi = 1$) $\Rightarrow G = D$. So in Friis, use $G_t, G_r = \text{these } D \text{ values (e.g. half-wave } G = 1.64)$.

ANTENNA AREAS (A_e, A_p)

Effective area <i>antenna's RX cross section; D from BEAM PARAMS or table above</i>
$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$
Physical cross section <i>wire dipole, diameter d; l see table above</i>
$A_p = l d \text{ (m}^2\text{), any dipole}$
Inverse: D from A_e
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)
Hertzian special case $D = 1.5$
$A_p^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119 \lambda^2 \text{ (m}^2\text{)}$
Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

FRIIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.

3. If TX is “half-wave dipole” use $G_t = 1.64$. 4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity ($G, D, P_r/P_t, \text{SNR}, \dots$):

$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}}$		$X_{\text{lin}} = 10^{X_{\text{dB}}/10}$	
dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2, 10 \text{ dB} \Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

NOISE & SNR

Noise power <i>thermal noise into matched receiver</i>
$P_N = k_B T_{\text{sys}} B$ (W)
$k_B = 1.38 \times 10^{-23} \text{ J/K}; T_{\text{sys}}$ system noise temp (K); B receiver bandwidth (Hz).
Signal-to-noise ratio
$\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs SNR > 10 dB for usable signal.

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m²); A_e effective area (m², $= \lambda^2 D / 4\pi$); β_{xz}, β_{yz} HPBWs in two planes (rad); ℓ, d aperture side / diameter (m).

Directivity (any shape, uniform illum.)
$D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) $\Rightarrow A_e \approx A_p$
Directivity from beamwidths <i>any aperture</i>
$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}}$ (use rad); circular: $D \approx 4\pi / \beta^2$

HPBW and D by aperture shape *uniform illum.; β in rad*

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88 \lambda / \ell_x$ $\beta_y \approx 0.88 \lambda / \ell_y$	$D \approx 4\pi / (\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88 \lambda / \ell$	$D \approx 4\pi / \beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. d)	$\pi d^2 / 4$	$\beta \approx 1.02 \lambda / d$	$D \approx 4\pi / \beta^2 = \pi^2 d^2 / \lambda^2$
Scaling rules $D \propto A_p / \lambda^2; \beta \propto \lambda / \sqrt{A_p}$			
Change	New D	New β	
Double ($A_p \rightarrow 2A_p$)	area $D_{\text{new}} = 2D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}} / \sqrt{2}$	
Double freq ($\lambda \rightarrow \lambda/2$)	$D_{\text{new}} = 4D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}} / 2$	